Literature Review: Machine Learning Approaches for Sports Match Outcome Prediction

Comparative Summary of Reviewed Studies (Refs 1–4 of 10)

Ref	Paper Title	Author(s)/Year	Methodology & Technology	Advantages	Limitations
1	**A Data- and Knowledge-Driven Framework for Developing Machine Learning Models to Predict Soccer Match Outcomes**	Berrar, Lopes & Dubitzky (2024)	Framework combining structured feature engineering with domain (football) knowledge; guides selection of ML algorithms (e.g., kNN, XGBoost) around engineered rating/recency features	Systematic, reusable methodology bridging data science and football expertise; improves interpretability and feature relevance over purely data-driven pipelines	Framework-level guidance, less focus on a single optimized model; still depends heavily on quality/availability of historical match data
2	**Incorporating Domain Knowledge in Machine Learning for Soccer Outcome Prediction**	Berrar, Lopes & Dubitzky (2019)	Recency- and rating-based feature engineering from the 2017 Soccer Prediction Challenge dataset; kNN and gradient-boosted trees (XGBoost) for 3-class outcome prediction	Domain-engineered features boosted accuracy over raw statistics; best-performing model (kNN) in the international prediction challenge, a strong benchmark result	Draws remained difficult to predict (~3-class accuracy notably lower than 2-class); feature engineering is labour-intensive and league/data-source specific
3	**A Predictive Analytics Framework for Forecasting Soccer Match Outcomes Using Machine Learning Models**	Wong, Li, Le, Bhanugu & Bhatia (2025)	Ensemble framework combining LightGBM (gradient boosting) with CNN-based feature extraction; model stacking/ensembling for outcome classification	Combines strengths of tree-based and deep-learning models for higher predictive accuracy; ensembling reduces variance and overfitting risk of any single model	Increased computational cost and pipeline complexity; ensemble reduces interpretability compared with single simpler models
4	**Predicting Football Match Outcomes: A Multilayer Perceptron Neural Network Model Based on Technical Statistics Indicators of the FIFA World Cup**	Luo, Quan & Cao (2025)	Multilayer Perceptron (MLP) neural network; trained on technical/tactical statistical indicators from FIFA World Cup matches	Captures non-linear relationships among technical performance indicators; tournament-specific tuning improves relevance to knockout/high-stakes matches	World-Cup-only data limits sample size and generalizability to league play; neural network results are less interpretable than statistical/tree models

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Comparative Summary of Reviewed Studies (Refs 5–7 of 10)

Ref	Paper Title	Author(s)/Year	Methodology & Technology	Advantages	Limitations
5	**From Players to Champions: A Generalizable Machine Learning Approach for Match Outcome Prediction with Insights from the FIFA World Cup**	Al-Bustami & Ghazal (2025)	Combines player-level and team-level feature sets; classification models trained/validated for generalizability across tournaments	Player-level granularity adds richer predictive signal beyond team aggregates; designed explicitly for cross-tournament generalizability	Requires detailed player-level data, which may be unavailable for lower-profile leagues; higher dimensionality increases risk of overfitting without careful regularization
6	**Sports Analytics and Machine Learning Approaches for Match Outcome Prediction**	Not listed in indexed sources (IJFANS, 2022)	Comparative study of Logistic Regression, Decision Tree, SVM, Random Forest, ANN, and XGBoost	Broad benchmarking helps identify best-performing algorithm class for the task; simple baseline models (e.g., logistic regression) offer interpretability	Comparative/benchmark focus rather than a novel optimized model; performance differences across algorithms may be dataset-specific and not generalize
7	**A Critical Comparison of Machine Learning Classifiers to Predict Match Outcomes in the NFL**	Beal, Norman & Ramchurn (2020)	Systematic comparison of 9 ML classifiers on NFL match data	Rigorous, critical benchmarking across many classifiers builds confidence in best choice; findings inform classifier selection for other American-football-style sports	Sport-specific (NFL) results may not transfer directly to soccer's low-scoring, draw-prone nature; no single classifier dominated across all evaluation metrics

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Comparative Summary of Reviewed Studies (Refs 8–10 of 10)

Ref	Paper Title	Author(s)/Year	Methodology & Technology	Advantages	Limitations
8	**Using Machine Learning and Candlestick Patterns to Predict the Outcomes of American Football Games**	Yu-Chia Hsu (2020)	Novel feature engineering using candlestick-pattern-style representations of game statistics; ML classifiers applied to these pattern-based features	Innovative feature representation borrowed from financial technical analysis; potentially captures momentum/trend patterns missed by conventional stats	Candlestick-pattern approach is unconventional and less validated in sports analytics literature; adaptation to soccer's continuous play is unproven
9	**Analysing Long Short Term Memory Models for Cricket Match Outcome Prediction**	Chakwate & Madhan (2020)	LSTM (recurrent neural network) architecture; trained on ball-by-ball cricket data to estimate dynamic win probability	Sequence modelling captures the evolving, ball-by-ball nature of match state; enables real-time, in-play win-probability updates	LSTM training is computationally expensive and needs large sequential datasets; sport-specific (ball-by-ball) design does not directly generalize to soccer's event structure
10	**Enhanced Cricket Match Prediction Using Kernel Methods for Feature Extraction and Back-Propagation Neural Networks**	Dhinakaran & Anbuchellan (2026)	Kernel-based feature extraction/selection; Back-propagation neural network (BPNN) for real-time ODI match outcome prediction	Kernel methods improve feature quality and reduce dimensionality before training; supports real-time prediction updates during ODI matches	BPNN prone to local minima and requires careful hyperparameter tuning; cricket-specific (ODI) design has limited direct transferability to soccer

Research Gaps and SDG Alignment

1. Identified Research Gaps

- **Limited cross-sport generalizability:** Models are tightly coupled to one sport's data structure (ball-by-ball cricket, NFL plays, soccer ratings); LSTM and candlestick-style features are rarely tested across sports.
- **Persistent difficulty predicting draws / low-margin outcomes:** Soccer's 3-class problem (win/draw/loss) shows lower draw accuracy than decisive results, and no study offers a targeted fix for this imbalance.
- **Interpretability vs. accuracy trade-off unresolved:** Ensemble/deep models (LightGBM+CNN, MLP, LSTM, BPNN) gain accuracy but lose interpretability; few pair high-capacity models with explainability methods (e.g., SHAP).
- **Data bias toward elite competitions:** Rich player/tournament-level data (FIFA World Cup, challenges) is unavailable for lower-profile leagues, limiting reach and equity of these methods.
- **No standardized benchmarking:** Studies use varying metrics/datasets, making it hard to judge whether an advantage is genuine or dataset-specific; a shared cross-sport benchmark suite is missing.
- **Real-time prediction underexplored outside cricket:** Live win-probability updating (LSTM, kernel BPNN) is mature mainly in cricket; equivalent in-play frameworks for soccer/football remain immature.
- **Computational cost vs. deployability:** Ensemble/deep architectures raise pipeline complexity, rarely evaluated against real-world deployment constraints (latency, cost, edge devices).

2. Sustainable Development Goals (SDG) Alignment

Although sports-prediction ML is not a traditional SDG-focused domain, its methodologies and applications intersect with several United Nations Sustainable Development Goals:

SDG	Goal	Relevance to This Research Area
SDG 4	Quality Education	Domain-knowledge-driven ML frameworks (e.g., Berrar et al.) offer reusable, interpretable methodologies that support data-science education at the intersection of sports and analytics.
SDG 8	Decent Work and Economic Growth	Predictive analytics underpins sports-technology and betting-integrity industries, creating specialized data-science jobs and supporting innovation-driven growth.
SDG 9	Industry, Innovation and Infrastructure	Ensemble, deep-learning, and kernel-based architectures reflect ongoing innovation in analytics infrastructure; benchmarking studies (NFL, cricket) strengthen that pipeline's rigor.
SDG 10	Reduced Inequalities	The data-availability gap between elite tournaments and lower-profile leagues is an equity concern; closing it would make predictive tools accessible across competition levels and regions.
SDG 17	Partnerships for the Goals	International challenges (e.g., the 2017 Soccer Prediction Challenge) exemplify open, collaborative benchmarking between academic and sporting communities.

References

1 Berrar, D., Lopes, P., & Dubitzky, W. (2024). A data- and knowledge-driven framework for developing machine learning models to predict soccer match outcomes. *[Journal/Conference not specified in source table]*.

2 Berrar, D., Lopes, P., & Dubitzky, W. (2019). Incorporating domain knowledge in machine learning for soccer outcome prediction. *Machine Learning*.

3 Wong, K., Li, T., Le, D., Bhanugu, S., & Bhatia, R. (2025). A predictive analytics framework for forecasting soccer match outcomes using machine learning models. *[Journal/Conference not specified in source table]*.

4 Luo, Y., Quan, S., & Cao, J. (2025). Predicting football match outcomes: A multilayer perceptron neural network model based on technical statistics indicators of the FIFA World Cup. *[Journal/Conference not specified in source table]*.

5 Al-Bustami, A., & Ghazal, T. (2025). From players to champions: A generalizable machine learning approach for match outcome prediction with insights from the FIFA World Cup. *[Journal/Conference not specified in source table]*.

6 Author(s) not listed in indexed sources. (2022). Sports analytics and machine learning approaches for match outcome prediction. *International Journal of Food and Nutritional Sciences (IJFANS)*.

7 Beal, R., Norman, T. J., & Ramchurn, S. D. (2020). A critical comparison of machine learning classifiers to predict match outcomes in the NFL. *[Journal/Conference not specified in source table]*.

8 Hsu, Y.-C. (2020). Using machine learning and candlestick patterns to predict the outcomes of American football games. *[Journal/Conference not specified in source table]*.

9 Chakwate, K., & Madhan, K. (2020). Analysing long short term memory models for cricket match outcome prediction. *[Journal/Conference not specified in source table]*.

10 Dhinakaran, D., & Anbuchellan, T. (2026). Enhanced cricket match prediction using kernel methods for feature extraction and back-propagation neural networks. *[Journal/Conference not specified in source table]*.

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